## TOPIC 1 Linear Equations and Inequalities

Name _____ Date _____

**Reminders**
**Words:** at least ≥ · at most ≤ · more than > · fewer than <.
**Solve like an equation:** undo + and −, then × and ÷.
**Flip the sign** when you multiply or divide both sides by a negative number.
**Graph:** ○ for < and >, ● for ≤ and ≥. Shade right for > and ≥, left for < and ≤.

**1.** What is the solution set for $-3x + 4 \geq 19$?

**Tip:** Dividing by a negative flips the sign.

**A.** $x \geq -5$

**B.** $x \leq 5$

**C.** $x \leq -5$

**2.** Solve the inequality, and graph the solution on the number line.

$5 - 2x < 11$

> **Tip:** Subtract 5, then divide by −2 and flip.

Subtract 5: $-2x <$ _____

Divide by −2 and flip: x _____ _____

**3.** A tire has 26 psi and loses 0.1 psi per minute. It must stay at or above 22 psi. Write and solve an inequality for the time $t$ in minutes.

> **Tip:** At or above → ≥. Divide by −0.1 and flip.

26 − 0.1t _____ 22

−0.1t ≥ _____

t _____ _____